



Automated Compliance Checking System for Structural Design Codes in a BIM Environment

Wonhui Goh^e, Jaeguk Jang^e, Sang I. Park^b, Bong-Hyuck Choi^f, Heuiseok Lim^d,
and Goangseup Zi^a

^aSchool of Civil, Environmental & Architectural Engineering, Korea University, Seoul 02841, Korea

^bResearch Institute for Safety Performance, Korea Authority of Land & Infrastructure Safety, Jinju 52856, Korea

^cKorean Construction Standards Center, Korea Institute of Civil Engineering and Building Technology, Goyang 10223, Korea

^dDept. of Computer Science and Engineering, Korea University, Seoul 02841, Korea

^eMember, School of Civil, Environmental & Architectural Engineering, Korea University, Seoul 02841, Korea

^fMember, Korean Construction Standards Center, Korea Institute of Civil Engineering and Building Technology, Goyang 10223, Korea

ARTICLE HISTORY

Received 30 May 2023

Revised 22 April 2024

Accepted 20 May 2024

Published Online 14 August 2024

KEYWORDS

Automated compliance checking (ACC)

Structural calculations

Building information modeling (BIM)

open BIM format

Industry foundation classes (IFC)

ABSTRACT

This study proposes a framework for developing an automated compliance checking (ACC) system that supports the verification of structural calculations in openBIM format. The framework consists of four main aspects: 1) classification of design codes depending on structural components and limit states to be checked; 2) preparation of a structural model according to Industry Foundation Classes; 3) interpretation of conditions in design codes into a machine-readable rule language; and 4) design of an ACC system, with a focus on the rule flow in the structural design code. The framework was demonstrated through a detailed example of an ACC system using ifcOWL, SWQL, and the Drools inference engine. The ACC system was developed to check the strength and serviceability of a prestressed concrete girder bridge and its components. The developed ACC system verified the actual design example. The proposed framework can help develop ACC systems in structural engineering and can be customized to meet user requirements and accommodate various data sources and design codes.

1. Introduction

Structural calculations should be verified with respect to design codes at every phase of the design process. The detailed results are conventionally displayed in the form of two-dimensional drawings. However, the huge dimensions and complicated details of civil structures often lead to conflict in construction work plans based on two-dimensional drawings (Leite, 2019). This would be one of the points to resolve for the BIM system introduced to construction engineering recently (Czmoch and Pękala, 2014).

Unlike a conventional two-dimensional drawing, a BIM represents structures in the form of objects in which various digital information is embedded (Kong et al., 2022). Once a BIM is built, various forms of data can be extracted to meet the needs of users, including conventional two-dimensional drawings (Kim et al., 2022). Quantitative information for structural calculations

can be mounted in a BIM for various applications. A BIM provides an adequate starting point for the development of an automated compliance checking (ACC) system for the quantitative verification of structural calculations (Teicholz et al., 2018).

Full-scale research on ACC in the architecture, engineering, and construction fields began in the 1960s. The initial concept was to develop a logical process for checking requirements in the design phase. Fenves (1966) proposed a compliance checking process for structural engineering that uses tabular decision logic to categorize design variables and values (Garrett and Fenves, 1987). Follow-up studies led to the development of the Standards Analysis, Synthesis, and Expression (SASE) system for comprehensively checking a structural design against related codes (Noland and Feng, 1971; Fenves et al., 1987; Garrett and Fenves, 1987). Compliance checking through comparison of regulations and standards with accumulated knowledge such as expert systems

CORRESPONDENCE Goangseup Zi ✉ g-zi@korea.ac.kr School of Civil, Environmental & Architectural Engineering, Korea University, Seoul 02841, Korea

© 2024 Korean Society of Civil Engineers

led to software development at the laboratory level (Eastman, 1981; Sriram et al., 1984; Cronenbold and Law, 1988; Rasdorf and Lakmazaheri, 1990; Yabuki and Law, 1993).

The rise of BIM intensified ACC research. BIM can serve as a good data source for ACC (Preidel and Borrmann, 2018). An ACC system can be applied to checking all activities related to the design, construction, and maintenance of a structure throughout its service life. This has increased demand for a model that can represent all necessary regulations and codes as well as reflect and apply data from multiple sources or software for different stakeholders (Amor and Dimyadi, 2021). OpenBIM is a software-independent model that can manage the geometric information of structural components as well as their use, control, and related attributes. It can also contribute to the organization and standardization of various data sources. Research on BIM-based ACC has been mainly based on the Industry Foundation Classes (IFC), which is a standard data schema for openBIM (Eastman, 2009; Eastman et al., 2009; Tan et al., 2010; Dimyadi and Amor, 2013; Zhang et al., 2013; Malsane et al., 2015; Dimyadi et al., 2016).

However, the use of ACC in structural engineering using openBIM is limited. Most tools developed for validating structural calculations are built using commercial software (Autodesk, 2022a; Bentley Systems, 2022; SkyCiv, 2022). These tools are software dependent, which hinders the incorporation of data from other sources. Users cannot see how codes are digitalized using these tools (Preidel and Borrmann, 2018; Amor and Dimyadi, 2021). These limitations make it difficult for users to follow rule processing in detail.

A few commercial software programs offer a transparent approach that allows users to understand how each code applies within rules. Solibri Model Checker (SMC) and Autodesk Navisworks are representative examples of such commercial software (Autodesk, 2022b; Solibri, 2022). These systems read the IFC model, standard data schema, and allow the user to adjust the parameters of the rules. However, the rules must be coded using functions provided by these systems. The definition of completely new rules within these systems suffers from limitations regarding national regulations and codes (Johannes et al., 2016; Preidel and Borrmann, 2016; Zhang et al., 2023).

Structural design codes often require cross-referencing (Rogers, 2023), leveraging results from one code within another. The current hierarchical structure of SMC restricts the results of a rule to only rules in the upper layer, hindering flexibility (Solibri, 2023). In Navisworks, using the results of one rule to check another simultaneously is challenging. A more flexible mechanism for rule execution is needed for ACC workflows in structural engineering.

Research on applying structural design codes to ACC is limited and concentrates only on applying a single structural member (Patlakas et al., 2018; Khattra et al., 2022). The complexity and computational intensity of structural design codes partially explain the limited use of ACC systems in structural engineering (Patlakas et al., 2018). Structural design codes classify structural members based on load (e.g., compression and shear) (KCSC,

2023), whereas BIM and IFC categorize parts into physical components. Additionally, structural components bear various load types simultaneously. Design codes should be selected appropriately considering these loads. Additional analysis is needed to bridge the gap between the structural design codes and BIM or IFC. The relationship between design variables and each code must also be considered. For example, a small change in the design value of an alignment-related variable will affect not only the bar spacing but also the shear strength and crack control. The relationships between variables and codes should be defined precisely, but this task becomes more complex when the scope expands to the whole structure.

The complexity extends to the process of preparing models for ACC. These models must represent various design variables for different codes, which requires extensions to the IFC data schema. Structural engineering research on BIM primarily focuses on integrating structural analysis models into IFC (Xu et al., 2019; Park et al., 2020; Sibenik and Kovacic, 2020; Jia et al., 2022). These efforts cannot fully address the needs of preparing models for ACC. The design variables for ACC and those used in structural analysis only partially overlap. ACC involves applying various load combinations to structural analysis and comparing only the maximum value of each with the standard value in the code. An ACC model also requires a data schema to define standard values that are used to determine whether each analysis result meets code requirements. Wu and Zhang (2022) proposed integrating building code variables into IFC for ACC, but research on extending variables specifically for structural design codes remains limited.

In this study, a framework is proposed for developing an ACC system to support the verification of structural calculations. The framework systematically applies the information of structural design codes to an ACC system in openBIM format, focusing on 1) selecting a design code according to the target structure, 2) implementing variables of the design code in IFC format, 3) writing conditions in complex structural code as rules, and 4) designing a mechanism for the ACC system to execute all rules simultaneously. This study covers design codes applied during design development in structural engineering. This paper presents the proposed framework, a sample application for the development of an ACC system for a prestressed concrete (PSC) girder bridge, and verification of the ACC system through its application to an actual design example.

2. Proposed Framework for ACC System Development

2.1 Digital Interpretation of Design Codes

Most requirements in structural design codes may be classified as optional or an inequality to satisfy depending on the conditions being considered:

$$Y = g(X), \quad (1)$$

$$f_R(X) \geq f_L(X), \quad (2)$$

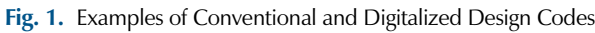


Figure 1 shows an example of how variables and conditions

are combined in a code on the minimum reinforcement of flexural members in the ultimate limit state. The minimum reinforcement area required by the code is compared with the calculated reinforcement area of a structural member. It has three requirements. The first requirement is of a similar type to Eq. (2). In this example, the condition is expressed in natural language, but it is more often expressed as an inequality. The second and third requirements, which concern the minimum reinforcement area, are calculated

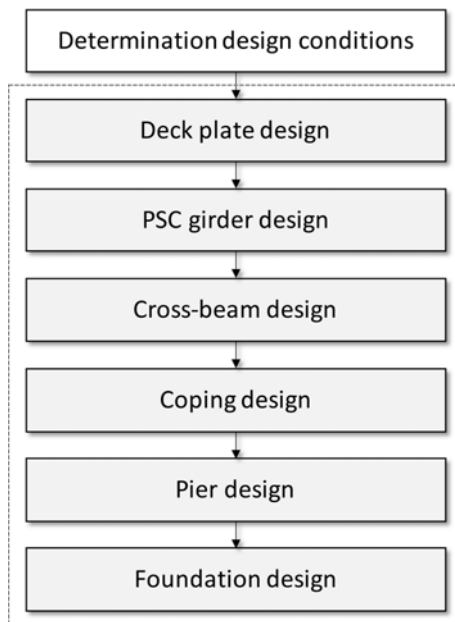


Fig. 3. Design Process for a PSC Girder Bridge

using Eqs. (6.3.1) and (6.3.2), respectively, as shown in Fig. 1. These follow the format of Eq. (1), and the greater of the two calculated values is selected for use. The design variables for each equation are listed. The conditions are defined as computer-readable rules, and the design variables are defined in IFC format.

Figure 2 shows the process of applying a design code to the ACC system. The information in the design code is classified as variables or conditions, which are then converted into a machine-readable format. Variables are defined in a BIM-based format, which allows structural calculations to be expressed in a model. Conditions are written as machine-readable rules, which are then checked against the model results.

Different design codes may be applicable to an ACC system depending on the target structure, which determines the scope. For example, Fig. 3 outlines the design process of a PSC girder bridge. The design conditions of the entire structure are identified. Then, each component is designed in top-down order from the superstructure to the substructure (e.g., foundation). Thus, the design process can be divided into groups of structural components.

Figure 4 shows the design process of a bridge deck: assume design values, calculate loads and parameters, and check whether conditions are satisfied. The design code referenced in each step is marked by a hatched box. The design code provides the conditions that the bridge deck design must meet. For structures designed according to the limit state method, such as a bridge, conditions may be grouped according to the limit state. In this case, the design code can be selected as follows: each component of a target structure is identified, and the relevant design codes are identified according to the limit states of these components.

The design codes used in this process can be applied to other structures. If the members of a structure are subjected to the same loads and have the same material structures, then the same codes

apply to those members. In other words, a structural component references multiple design codes, and a design code may reference multiple members. The relationship between design codes and structural components can be analyzed through the structural design process.

2.2 Implementation of Design Information in Digital Format

A BIM-based ACC system uses 3D digital models as the source and design codes for as the target of the compliance check. The models should represent all necessary codes (Fenves et al., 1995), but IFC schema cover very limited information related to design codes (Zhang and El-Gohary, 2016). To add the variables in a design code to an ACC system, they need to be defined in the IFC schema. New attributes can be added to the IFC in two ways: defining them as new entities or defining them through property sets (PSets). The first approach of adding a new entity deforms the IFC information structure, which makes it difficult to use the IFC file in other applications. In contrast, using a PSet to add a new attribute does not change the structure because it uses entities already defined in the existing IFC. PSets also allow data to be processed independently of the IFC version of the structural model. Therefore, the second approach of using PSets was selected in this study.

Another challenge for ACC systems is accessing the objects required by the design code from the IFC model (Amor and Dimyadi, 2021). This is generally addressed by formalizing ontology-based rules for extracting the required information (Zhou and El-Gohary, 2016; Jiang et al., 2018; Zhou and El-Gohary, 2018; Sydora and Stroulia, 2020; Zhou and El-Gohary, 2021). The ontology is specialized for queries used to find necessary information in a dataset. An ontology representing the IFC dataset is needed. The Resource Description Framework (RDF) query language can be used to express an ontology and has a semantic structure similar to that of IFC. RDF graph data structures are defined according to subject, object, and subject-object relationships while IFC data structures are defined according to entities, properties within entities, and relationships between entities. RDF graphs can be applied to converting IFC to semantic language for application of other software tools. Several converters are applied to convert IFC files to RDF graphs. In this study, the Ugent-Aalto IFC to RDF converter was used (Pauwels and Terkaj, 2016), which converts IFC to Terse RDF Triple Language (Turtle) (Beckett et al., 2022). The conversion from IFC to RDF graph is called ifcOWL, and it is used to check structural calculations. Fig. 6 shows an example of the converted structures, where the entities of IFC are converted by ifcOWL to individuals.

2.3 Rule Interpretation

Transforming conditions in the design codes from natural language into machine-readable rules is a very important task for the development of an ACC system. Expert knowledge is needed for accurate interpretation of the design code. Rules are used to extract entities from the model and check their values for compliance. The scope of the design code has to be considered;

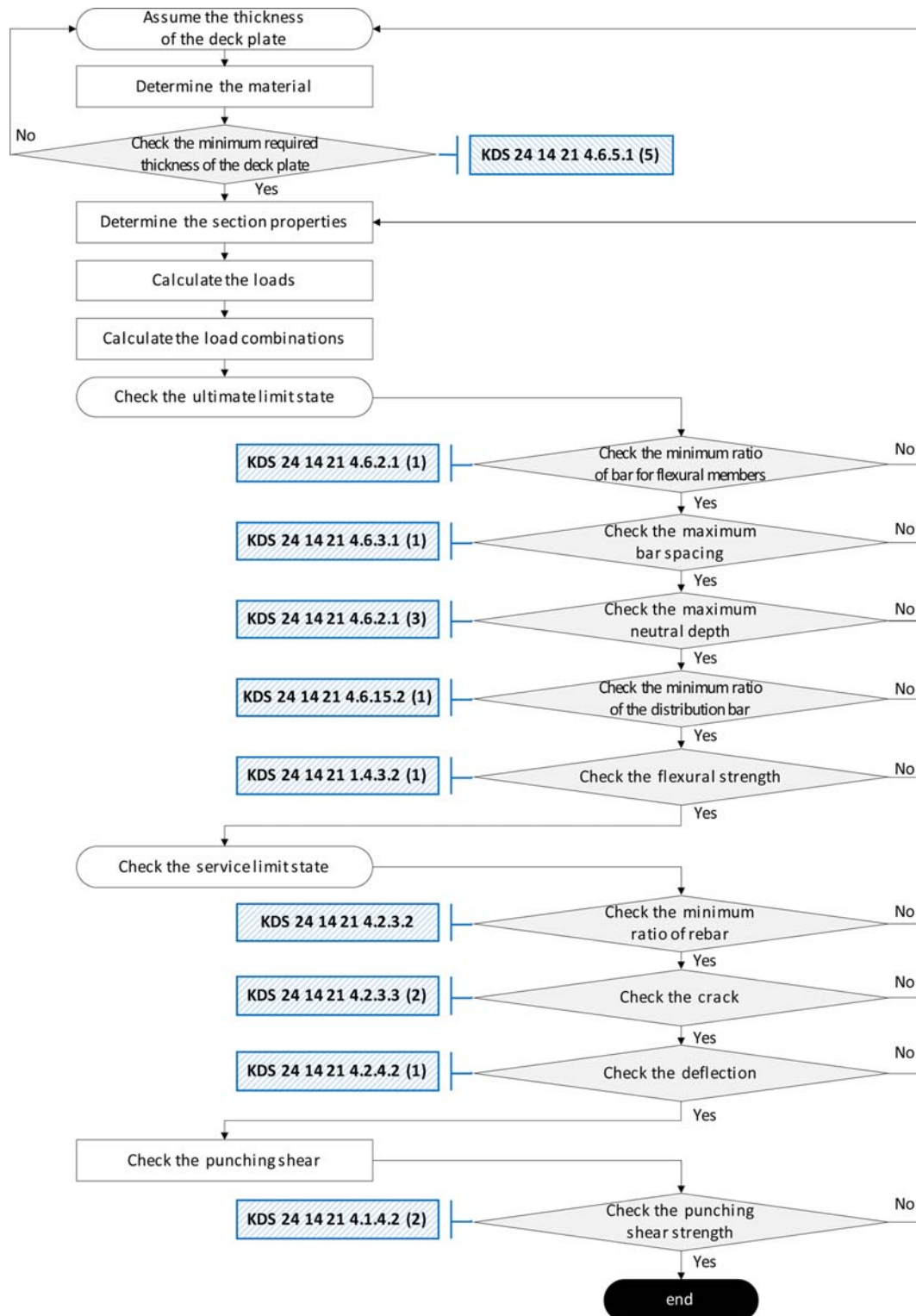


Fig. 4. Design Process of the Bridge Deck

for example, one code may be applicable to the entire structure while another is only applicable to each member. The logical process and exceptions in the code should be identified. Design codes have many exceptions and a complicated comparison logic, so understanding the logical flow is very important.

Next, a language should be chosen for writing the rules. A design code can be converted into a machine-readable form in various ways. As noted earlier, ontology-based rules are used to access data in a model. Rule languages are similar in form to natural language, so they are easily encoded by domain experts

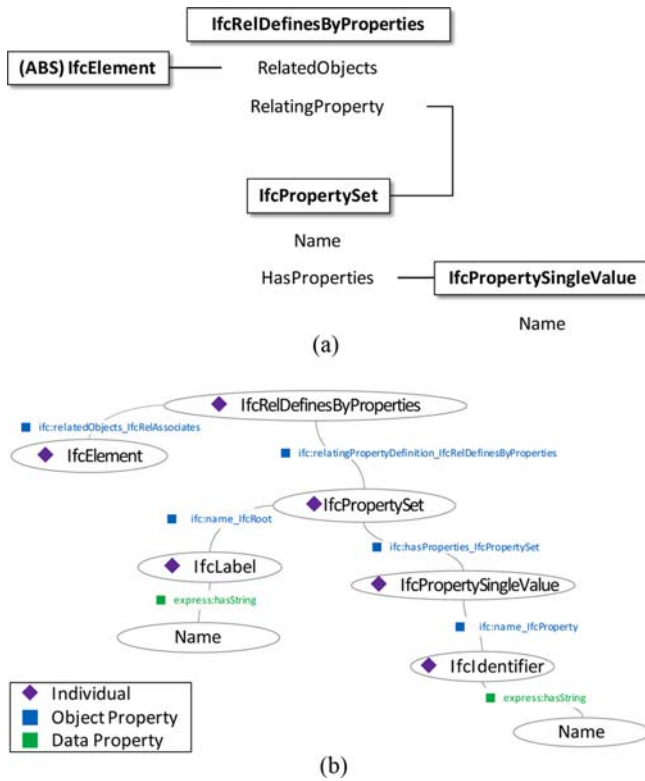


Fig. 5. Examples of Data Structures: (a) IFC, (b) ifcOWL

(Jiang et al., 2022). Semantic Web Rule Language (SWRL) can be used to express rules for an ontology (Horrocks et al., 2022). SWRL is used to write rules that can infer new knowledge from existing ontologies. In a BIM environment, SWRL infers whether design variables defined by an ontology (ifcOWL) meet relevant code conditions and derives new knowledge, such as review

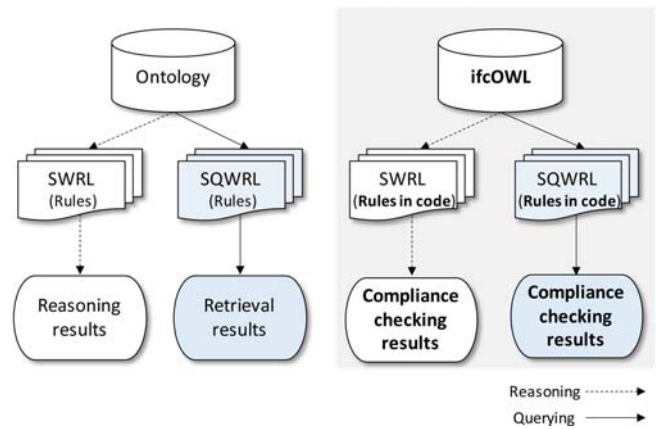


Fig. 6. Comparison of SWRL and SQWRL, and Their Application to the ACC System

results. Many researchers have used SWRL to write design code conditions as machine-readable rules (Zhong et al., 2012; Zhang et al., 2015; Jiang et al., 2018).

SQWRL is a SWRL-based language with an enhanced query function (O'Connor and Das, 2009). Both SWRL and SQWRL consist of two main parts: the antecedent (also known as the body) and the consequent (also known as the head). The antecedent is the “if” part of the rule. It contains one or more atoms that represent conditions. The consequent is the “then” part of the rule. It describes the inference or conclusion obtained when the conditions in the antecedent are met.

In the ACC system, the antecedent conducts compliance checking, whereas the consequent processes the results. SWRL and SQWRL differ in the implementation of the consequent, which handles the processing of results. For both SWRL and SQWRL in the ACC system, the antecedent involves the extraction of the

7.2.1 Shear strength

- (1) Except for members which are designed in accordance with Appendix I, cross sections subjected to shear shall be designed by Eq. (7.2.1).

$$V_u \leq \phi V_n \quad (7.2.1)$$

where V_u is the factored shear force at the considered cross section, and V_n is the nominal shear strength calculated by Eq. (7.2.2).

$$V_n = V_c + V_s \quad (7.2.2)$$

where V_c is the nominal shear strength provided by concrete calculated in accordance with 7.3 or 7.12, and V_s is the nominal shear strength provided by shear reinforcement in accordance with 7.4, 7.10.2(5), or 7.12.

- Code related to review
- Code related to calculation
- Code referenced

Fig. 7. Analysis of Types and Referencing Relationships of Structural Design Codes

design value from ifcOWL and verifying whether the design information in ifcOWL satisfies the conditions. Depending on whether SWRL or SQWRL is used, the review result processed in the antecedent is handled differently in the consequent. For instance, SWRL defines the consequent to store new knowledge, such as review results, in ifcOWL. SQWRL defines the consequent to display the results in the output window.

Using SWRL or SQWRL to write the ontology rules allows information to be automatically derived through reasoning and querying. As shown in Fig. 6, if conditions in the code are written in rule language and are applied to the ifcOWL ontology, both methods are applicable. In SWRL, the condition is written in the antecedent, and storing the check result in ifcOWL is written in the consequent. In SQWRL, the condition is written in the antecedent, but only the output of the check result is written in the consequent. In other words, SWRL stores the check result in ifcOWL, but SQWRL simply outputs the check result for ifcOWL. SQWRL requires additional work to save the check results in ifcOWL. However, using SWRL requires a longer computation time than using SQWRL. Thus, SQWRL was selected in this study.

2.4 Process of Compliance Checking

Most items in a structural design code may be classified as review or calculation. Fig. 7 shows the design code used to verify the shear strength of structural components (KCI, 2012). In Fig. 7, Eq. (7.2.1), highlighted in red, is associated with review-related codes, whereas Eq. (7.2.2), highlighted in blue, pertains to calculations for review. Review-related codes obtain their results by referencing calculation outcomes. Additionally, the variables in Eq. (7.2.2) are derived from equations in other codes (e.g., 7.3,

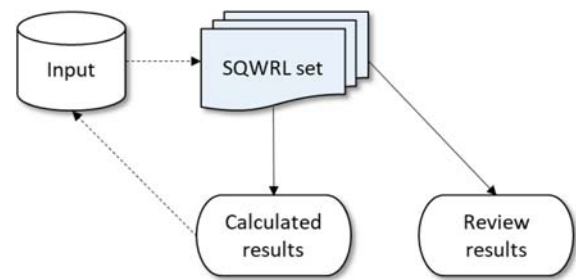


Fig. 8. Input and Output Structures for Rule Execution

7.12, and 7.4), which are highlighted in green. Therefore, the referencing relationships between these codes should be analyzed to ascertain the appropriate execution order of the rules. Our system architecture is designed to use the results of one code within another. The obtained results can then be used as an input for another SQWRL set, as demonstrated in Fig. 8. By examining the relationship between the variables used in the rules, we can establish the order for executing SQWRL.

Ontology-based ACC systems generally perform compliance checking through three main functions: an application programming interface (API) for managing and updating the ontology model, an API for rule management, and an inference engine based on the model and rules. OWL API (Horridge and Bechhofer, 2011) is an open-source tool for ontology model management that can be used to load, manipulate, and save ontology files. An ACC system can use this API to load ifcOWL models containing information about structural calculations. The SWRL API can be used for rule management. This API enables SQWRL rules to be processed and manages all rules for checking the conditions in design codes. The compliance checking process reviews rules

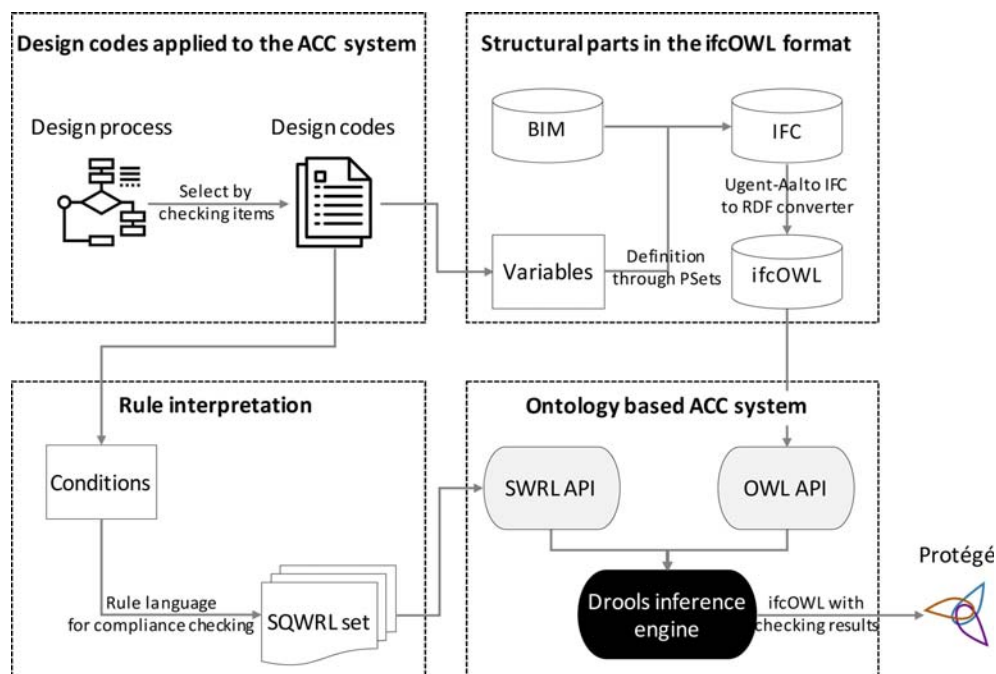


Fig. 9. Interpretation of the Design Process, Augmentation of IFC, Implementation of Rules, and Ontology-Based ACC

based on calculation and comparison functions built into the SWRL API. The open-source Drools engine can be utilized as the inference engine, which is the most important part of the ACC system (Proctor, 2011). The inference engine generates new facts from existing facts based on the stored rules. The engine makes inferences and provides conclusions based on the semantic relationships and rules in the ontology. In other words, SQWRL is executed sequentially using the Drools engine, with individuals being created via the OWL API following each SQWRL execution.

2.5 Framework

As shown in Fig. 9, the proposed framework for developing an ACC system comprises four stages. First, the design process is extracted from actual structural calculations. Items to check against the design code at each step are identified. Second, the design code is transformed into a digital format, where the code is interpreted as a combination of variables and conditions. Third, the data including the variables from the design code and digital information of the structure are converted into the IFC data format, which is transformed further to ifcOWL. The conditions from the design code are written in a machine-readable rule language such as SQWRL. Finally, an inference engine such as Drools is applied to the data and rules. The results of the compliance check can be reviewed by using the ontology editing software Protégé.

3. Development of ACC for PSC Girder Bridge

3.1 Target Structure

The proposed framework was applied to developing an ACC system, which was then implemented in a case study. A PSC girder bridge has a simple design and low construction cost, so it is often applied to single spans. Thus, a PSC girder bridge was selected as the target structure. The structure of a PSC girder bridge can be divided into a superstructure and substructure. An

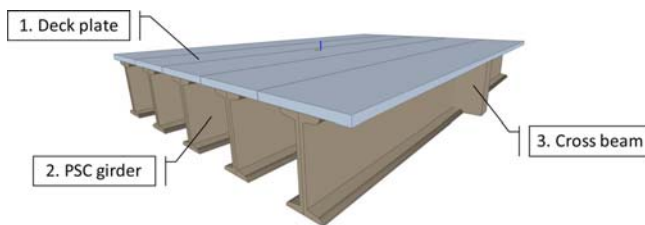


Fig. 10. Element of the PSC Girder Bridge Superstructure

Table 1. Level of Detail in a BIM by Design Phase

Level of detail (LOD)	Design phase
LOD 100	Conceptual design
LOD 200	Schematic design
LOD 300	Design development (low)
LOD 350	Design development (high)
LOD 400	Construction
LOD 500	As-built

ACC system was developed for the superstructure, which receives loads directly. As shown in Fig. 10, the following components of the superstructure design were considered for the case study: the deck plate, PSC girder, and crossbeam. According to the BIM basic guidelines (KMOLIT, 2020), the level of detail (LOD) can be defined according to the design phase, as summarized in Table 1. For example, the LOD is 100 for the conceptual design phase, 400 for the construction phase, and 500 for the as-built condition. In this study, the LOD of the structural model for the compliance check was set to 350. This corresponds to the design development (high) phase and includes the design of reinforcing bars.

3.2 Applied Design Codes

The applicable design codes for checking the superstructure design of the PSC girder bridge were identified. First, the necessary conditions for each component (i.e., deck plate, PSC girder, and crossbeam) to satisfy were identified. The ultimate and serviceable limit states of each component were identified, and the related design codes were identified. The contents of the design codes were then checked for whether they were applicable to the ACC system. Design codes of uncertain contents (e.g., sufficient strength) were excluded from the scope. In addition, the design code related to the strut strength was excluded from the scope. Table 2 summarizes the checked items for each component and the applied design codes.

3.3 Conversion of Design Information into IFC Format

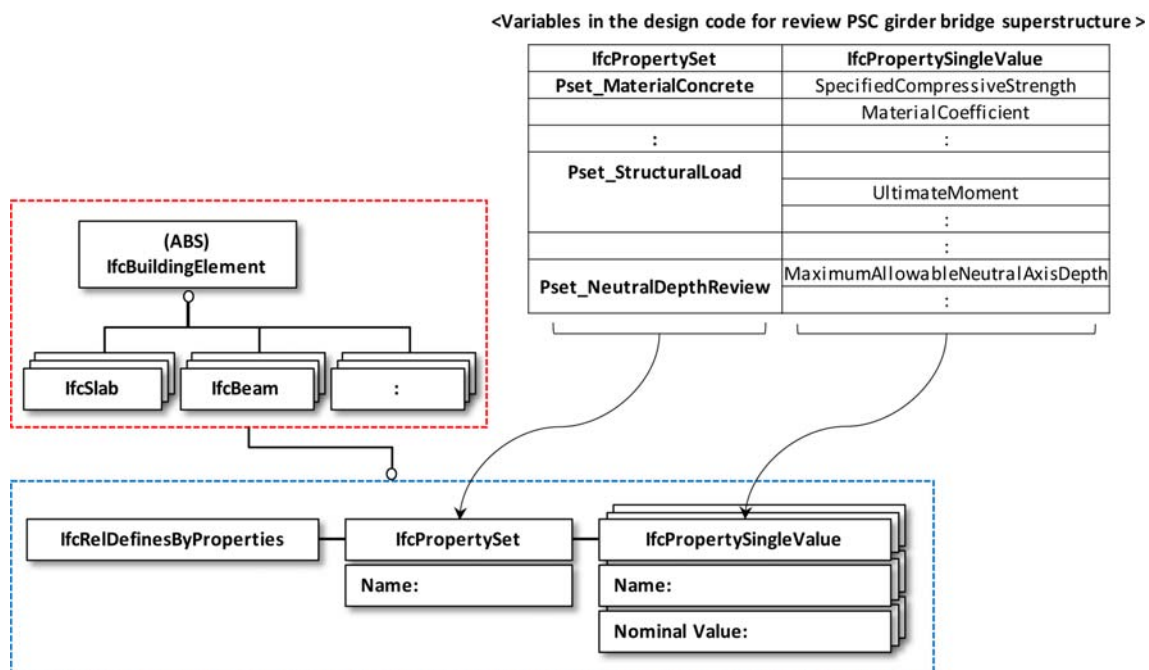
As shown in Fig. 11, the variables were divided into a two-level hierarchy to be defined in the IFC schema using PSets: the upper level is *IfcPropertySet*, and the lower level is *IfcPropertySingleValue*. The resource layer of the IFC data schema was referenced to design the variable hierarchy. The resource layer is one of the four layers constituting the IFC, and its data are generally used to express information of a component. Therefore, data defined in the resource layer can be considered analogous to a variable in a design code. The resource layer has several entity types, among which the material/profile/structural load was selected to represent variables in design codes. These entities can be subdivided; for example, material entities can be subdivided into concrete, steel, and tendon materials. These entities were defined as *IfcPropertySet*. The variables belonging to an *IfcPropertySet* were defined as *IfcPropertySingleValue*. Each checked item was also defined as an *IfcPropertySet* (e.g., *PSet-NeutraldepthReview*) to store the results. PSets were added for relevant component. Table 3 lists the IFC data of the concrete materials. PSet is connected to *IfcSlab*, which is used to define a deck plate. *IfcPropertySet* of concrete is defined as *Pset-MaterialConcrete*, and the compressive strength or strain limit for concrete is defined as an *IfcPropertySingleValue*. The design values of the structure were entered in the IFC, which was converted to an ifcOWL and loaded into the ACC system.

3.4 Rule Structure and Execution Logic

In SQWRL, the rules mainly comprise defining variables, checking them for compliance, and outputting the results. The variable

Table 2. Checked Items for Each Member of the PSC Girder Bridge Superstructure and the Related Design Code

Components	Limit states	Items	Design codes	Remarks
Deck plate	Ultimate limit state	Minimum required thickness of deck plate	KDS 24 14 21 4.6.5.1 (5)	Excluded
		Minimum ratio of bar for flexural members	KDS 24 14 21 4.6.2.1 (1)	
		Maximum spacing of bar	KDS 24 14 21 4.6.3.1 (1)	
		Maximum neutral axis depth	KDS 24 14 21 4.6.2.1 (3)	
		Minimum ratio of distribution bar	KDS 24 14 21 4.6.15.2 (1)	
		Minimum flexural strength	KDS 24 14 21 1.4.3.2 (1)	
	Service limit state	Minimum ratio of bar	KDS 24 14 21 4.2.3.2	
		Crack	KDS 24 14 21 4.2.3.3 (2)	
	Punching shear	Deflection	KDS 24 14 21 4.2.4.2 (1)	
PSC girder	Ultimate limit state	Punching shear strength	KDS 24 14 21 4.1.4.2 (2)	Excluded
		Minimum ratio of bar for flexural members	KDS 24 14 21 4.6.2.1 (1)	
		Minimum ratio of bar for brittle failure	KDS 24 14 21 4.1.1.2 (7)	
		Minimum flexural strength	KDS 24 14 21 1.4.3.2 (1)	
		Minimum shear strength	KDS 24 14 21 4.1.2.1 (1)	
		Minimum interface shear strength	KDS 24 14 21 4.1.2.5 (1)	
	Service limit state	Strength in Strut-Tie model	KDS 24 14 21 4.1.5.2	
		Maximum tendon load	KDS 24 14 21 1.5.7.2 (1)	
		Maximum concrete stress at tendon	KDS 24 14 21 1.5.7.2 (2)	
Crossbeam	Ultimate limit state	Maximum stress	KDS 24 14 21 1.4.3.3 (3)	Excluded
		Deflection	KDS 24 14 21 4.2.4.2 (1)	
		Minimum ratio of bar for flexural members	KDS 24 14 21 4.6.2.1 (1)	
		Maximum neutral axis depth	KDS 24 14 21 4.6.2.1 (3)	
		Minimum flexural strength	KDS 24 14 21 1.4.3.2 (1)	
		Punching shear strength	KDS 24 14 21 4.1.2.1 (1)	
		Maximum spacing of the distribution bar	KDS 24 14 21 4.6.3.2 (3)	Excluded

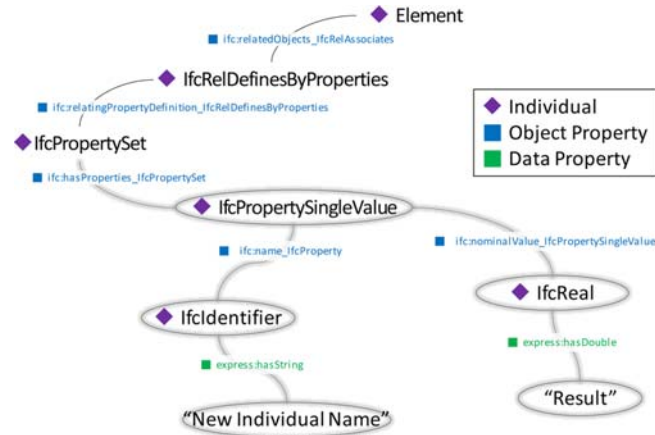
**Fig. 11.** Layering Variables within Design Codes and IFC Schema Definition Through PSet

definition and check are written in the antecedent, while outputting the results is written in the consequent. The variable definition includes the design variables of the structure in ifcOWL. The

compliance check defines formulas for checking conditions. The final parts outputs the resulting values and the location where they are stored. The location function are defined to store the

Table 3. IFC Data Example of PSet for Design Code Variables

```
#135=IFCSLAB('3$Y1qZFGzBO9A70r9J2$f$',#42,'X2\BC14B2E5\X0\slab_240T:378778',$,'X2\BC14B2E5\X0\slab_240T',#119,#134,'378778',FLOOR.);
#114000= IFCRELDEFINESBYPROPERTIES('4gfVO40P5EfQyKZ_bf0118',#42,$,$,(#135),#114001);
#114001= IFCPROPERTYSET('4aRZfLiLD9wasofRec119',#42,'Pset_MaterialConcrete',$,(#114002,#114003,#114004,#114005,#114006));
#114002= IFCPROPERTYSINGLEVALUE('MaterialConcreteCompressiveStrength',$,IFCREAL(35.0),$);
#114003= IFCPROPERTYSINGLEVALUE('MaterialConcreteLimitedStrain',$,$,$);
#114004= IFCPROPERTYSINGLEVALUE('MaterialConcreteCrackingTensileStrength',$,IFCREAL(3.45),$);
#114005= IFCPROPERTYSINGLEVALUE('MaterialConcreteTensileStrength',$,IFCREAL(2.415),$);
#114006= IFCPROPERTYSINGLEVALUE('MaterialConcreteCoefficient',$,IFCREAL(0.65),$);
```

**Fig. 12.** Individuals and Properties Created after Rule Execution

resulting value and storage location together. SQWRL is executed in sequence, and individuals are created after each SQWRL execution to save the check results. An algorithm was developed to save the SQWRL execution results. The newly created individual is defined as an *IfcPropertySingleValue* in the lower-level *IfcPropertySet* in IFC. Therefore, the saved location is based on the location of *IfcPropertySet*. As shown in Fig. 12, individuals and object and data properties were created based on the location. The gray areas in this figure indicate individuals and properties that were newly added after rule execution. *IfcPropertySingleValue*, *IfcIdentifier*, and *IfcReal* Individual were created from the starting point, and an object property that connects each individual was created as well. In addition, creating a data property stores the variable name and resulting value in *IfcIdentifier* and *IfcLabel*, respectively.

SQWRL has an if-then (i.e., first-order logic) structure, so if the antecedent is satisfied, information about the consequent is

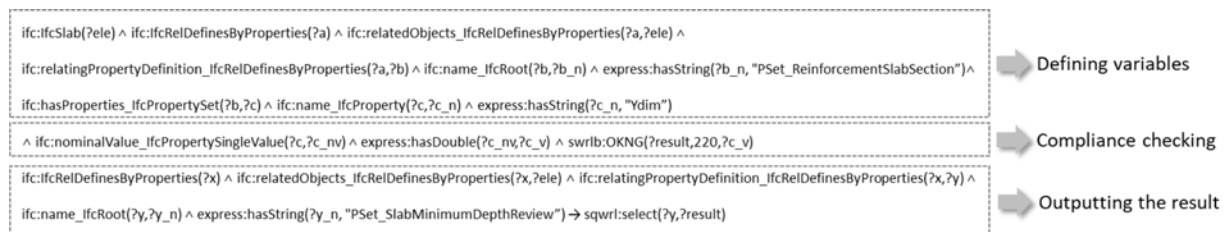
Table 4. Description of the Check Function

Expression	Description
swrlb:OKNG(?result,?var1,?var2)	var1 < var2: result = "OK"
	var1 ≥ var2: result = "NG"

output. Therefore, conditions are classified into two rules for writing if-else logic statements. Rules with OK and NG results should be written separately. SQWRL has various built-in comparison and mathematical functions, and the user can define additional functions. In this study, a checking function was defined for processing if-else logic statements so that conditions with OK and NG results can be checked according to one rule. As presented in Table 4, if the values of var1 and var2 are defined, OK is when var1 is less than var2, and NG is when var1 is greater than or equal to var2. For example, var1 may be the load applied to a structure and var2 may be the strength of the structure. If the load is less than the strength, then OK is obtained. If the load is greater than or equal to the strength then, NG is obtained.

Figure 13 summarizes an example rule of compliance checking. The design code is only applicable to the deck plate. The condition is that the height of the deck plate must be 220 mm or more. The rules are divided into three parts, as shown in Fig. 13: defining the design variables applied to the conditions, compliance checking, and outputting of results. The variables are defined to extract the Ydim value linked to the individual in the *IfcSlab* class representing the deck plate. Then, this value is checked for compliance using the check function. Finally, the storage location and resulting value of the new individual are output in the same manner as the calculation rule.

The execution order is determined by analyzing the referencing relationships between the codes summarized in Table 2. In

**Fig. 13.** Example of Converting Design Code Conditions into Rules

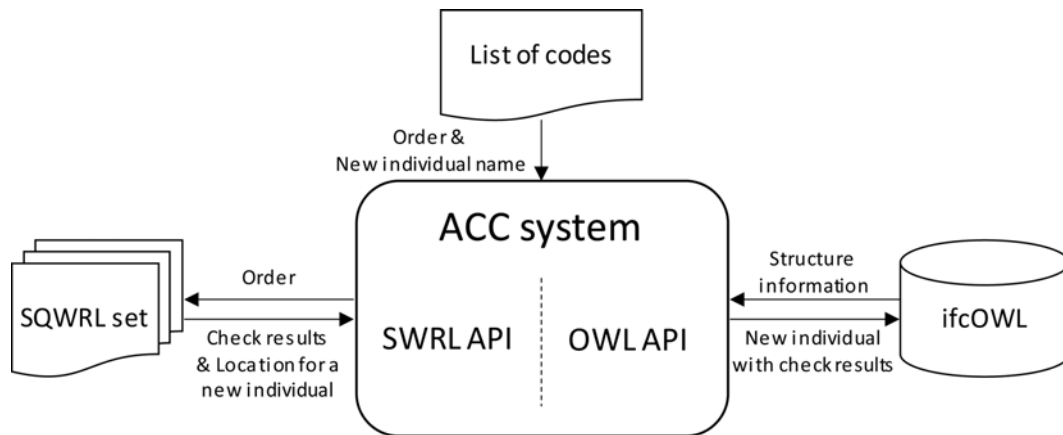


Fig. 14. Process of the ACC System

structural design codes, the calculation outcomes often serve as prerequisites for reviewing other codes. Therefore, the execution order of SQWRL rules is set to execute the rules for calculation first, followed by the rules for review. Moreover, some rules for calculation depend on the results of prior calculations. The execution order is thus established by considering the interdependencies between the results of various calculations. In the SWRL API, SQWRL rules are stored to execute all rules sequentially in a single process.

3.5 ACC System Architecture

The ACC system was developed in a Java environment by using the editing software Eclipse. The ACC system was implemented by using the open-source OWL API and SWRL API to execute the SQWRL rules in the loaded ifcOWL and output the results of the compliance check. As shown in Fig. 14, the ACC system requires three input files: a list of design codes, an SQWRL set of rules, and ifcOWL. The compliance checking process is repeated performed until all rules stored in the list of codes have been executed.

4. Application of Structural Calculation to ACC

4.1 Case Study: Structural Calculation on Dally Bridge

The Dally Bridge in Mokpo, South Korea was selected as a case study. This is a three-span continuous PSC girder bridge comprising a two-lane road and sidewalks. The superstructure comprises a reinforced concrete (RC) deck, PSC girder, and RC crossbeams, and the substructure comprises T-shaped piers and a direct foundation. The bridge was designed according to the process described in Figs. 3 and 4. The applied vehicle live load was set to KL-510 (KMOLIT, 2016), which corresponds to a first-class bridge load condition. The design was based on the limit state method.

A BIM for the Dally Bridge was created by using the program Revit 2020, where only the position and shape of the components were defined. The model was extracted by using IFC4 Reference View, and the shape was checked by using the IFC viewer

program. Information necessary for the compliance check was manually entered into the extracted IFC file, which was then converted to ifcOWL through the Ugent–Aalto IFC to RDF converter. This completed the preparation of the structural model for the compliance check. The structural information of the bridge was saved in ifcOWL, and all information necessary for the compliance check was prepared.

4.2 Results

The design of the Dally Bridge was checked for 12 items, as summarized in Table 2: nine for the deck plate, seven for the PSC girder, and three for the crossbeam. ifcOWL, which stored the data for each checked item, was loaded into the ACC system, and the results of the compliance check were saved in the updated ifcOWL and checked in Protégé. Fig. 15 shows the compliance check results for the minimum bar ratio of the deck plate. *IfcPropertySingleValue-900356* was connected to *IfcIdentifier-900356*, which had the name *FlexuralMemberMinRebarAreaReviewResult*. In addition, *IfcPropertySingleValue-900356* individual was connected to *IfcLabel-900356* individual, which had a result of OK. The compliance check results were examined by arbitrarily changing the section properties. The minimum bar ratio for flexural members in the ultimate limit state was checked by changing the amount of tensile reinforcing bars. The conditions of the compliance check are given in Equation:

$$A_{s,min} \geq \max\left(\frac{0.25\sqrt{f_{ck}}b_w d}{f_y}, \frac{1.4}{f_y}b_w d\right), \quad (3)$$

where f_{ck} is the concrete compressive strength, f_y is the reinforcement strength, b_w is the section width, and d is the effective depth. Dally Bridge had $f_{ck} = 30$ MPa, $f_y = 400$ MPa, $b_w = 1,000$ mm, and $d = 180$ mm. Thus, the amount of reinforcing bars should be greater than or equal to 616.2 mm^2 . The existing amount of 1986.0 mm^2 was changed to 500 mm^2 . The changed data was applied to the ACC system, which output an NG result for the compliance check. In addition, the other checked items related to the amount of tensile bars also output NG results, such as the minimum amount of bars for flexural members and cracking in the service limit state.

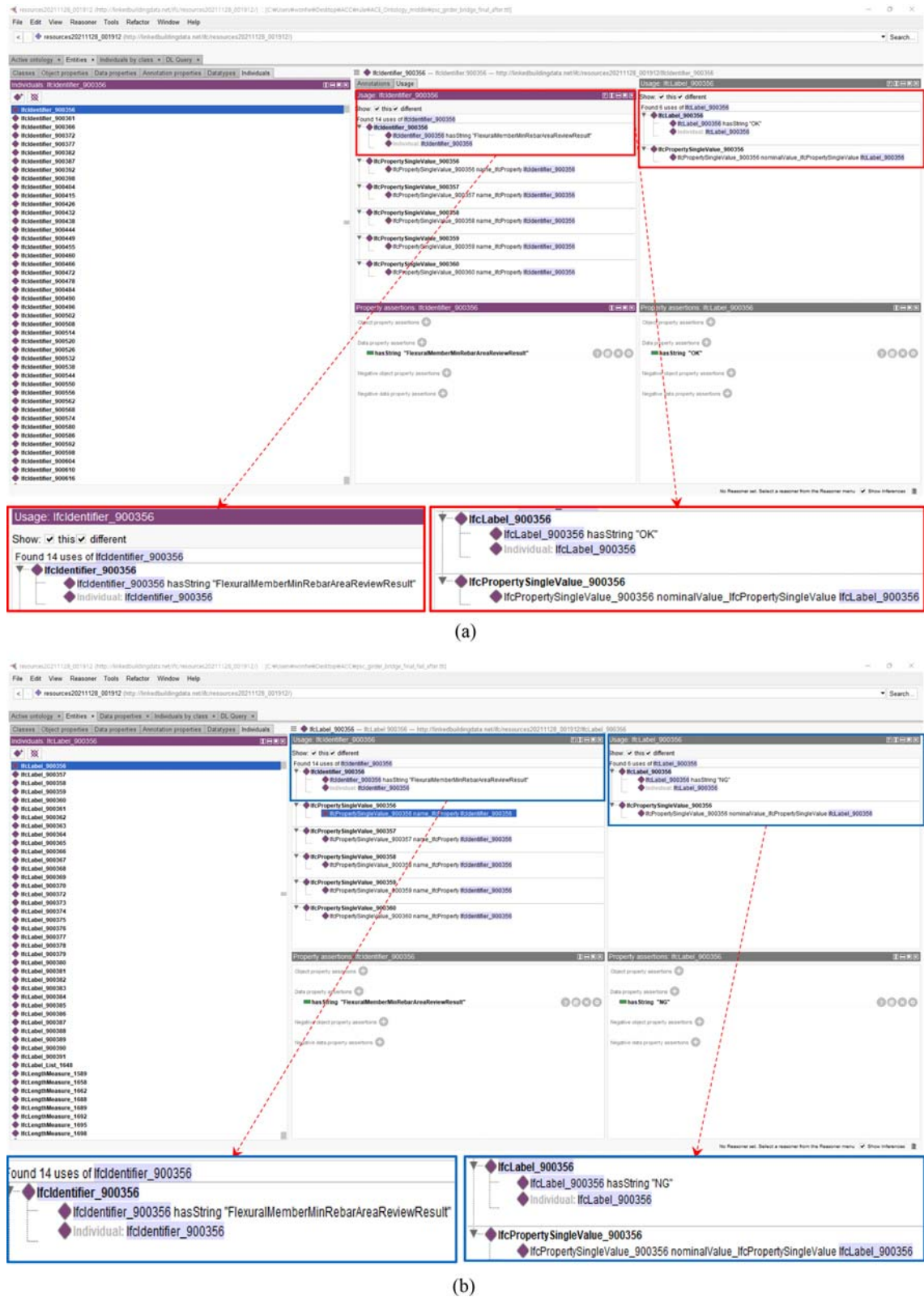


Fig. 15. ACC System Execution Result: (a) Actual Design, (b) Modified Design

4.3 Discussion

The ACC system was developed and verified using a PSC girder bridge as the target structure. Its applicability was further confirmed

by applying the system to Dally bridge in South Korea, a PSC girder design example, and verifying the review results. The proposed research methodology has two main features.

First, a structural design process was employed to determine the design codes applicable to the ACC system. Applying structural design codes to the ACC system required additional efforts to classify the codes for each component of the target structure. This necessity arose because structural components are classified based on loads in structural design codes, but they are classified based on physical components in BIM or IFC. The design process in this study effectively bridges this gap. For the studied PSC girder, the main components (deck plate, girder, and crossbeam) were identified, and appropriate design codes were selected for each component.

Second, the execution order of the structural design codes was determined by identifying the referencing relationships between the codes. The ACC system is also designed to support mechanisms for using the results of one code in another within a single process. The standard values required for review were calculated and compared with the design values to derive the review results. Additionally, the variables constituting these standard values are often determined by referencing other codes, which underscores the strong cross-referencing nature of structural design. Thus, the interdependencies of the codes were analyzed, and the execution sequence was determined accordingly. For example, the execution order for the codes applied to the PSC girder bridge was identified. The mechanism of the ACC system, which handles calculations and reviews simultaneously, enables the concurrent performance of these tasks. This was verified through a comprehensive design review of the Dally Bridge.

5. Conclusions

In this study, a framework is proposed for developing an ACC system to incorporate structural design codes. A design code can be represented as conditions and variables, which can then be converted into rules compatible with IFC. The proposed framework was verified through its application to the design of the superstructure of a PSC girder bridge. The main conclusions are as follows:

1. The design code for the target structure of the ACC system was analyzed through the structural design process. Structural design codes classify structural members based on their load, whereas BIM or IFC categorizes parts based on physical components. The structural design process helps resolve these differences. The conditions that the structure must meet are extracted from the design codes.
2. The design code information can be classified as variables and conditions. These variables can then be defined using PSets without modifying the IFC data structure. This enables compatibility with various commercial software. This approach enables the application of structural calculation values to the IFC model of the ACC system. The conditions of the design code are converted into rules using SQWRL, and a new function is defined so that a single SQWRL rule can process if else logic simultaneously.
3. The execution order of SQWRL rules is determined by considering the interdependencies between codes. To facilitate the processing of rules for calculation and review in a single process, the compliance-checking mechanism in the ACC system was designed to use the results of one code within another.
4. The ACC system was developed for the superstructure design of a PSC girder bridge. The structural calculation was converted into ifcOWL format, and the ACC system used SQWRL rules to check the actual design value in ifcOWL. The system successfully verified the strength and serviceability of each component, thus confirming the applicability of structural design codes to the ACC system.

We attempted to establish a general framework for developing ACC systems in structural engineering. However, a challenge arose when the target structure of the ACC system was expanded. In this field, design codes are interconnected and must follow a specific execution sequence. Expanding the scope of the target structure for the ACC system complicates the analysis of this execution sequence for all codes. Several independent yet interconnected sets of rules should be defined to tackle this issue.

The primary function of this ACC system is to ensure that structural designs conform to safety regulations and comply with structural design codes. With this ACC systems, structural engineers can identify the limits set by these codes and explore various designs within these boundaries. Designs can be automatically generated within this ACC system, facilitating the use of various optimal designs. These designs can prioritize factors such as cost efficiency and carbon emission reduction. Furthermore, the ACC system offers the capability to compare the boundaries of design codes across different countries for the same structural components.

Acknowledgments

The authors are grateful to Seongwon Jung for helping to implement the ACC system. This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (No. NRF-2021R1A5A1032433 and NRF-2023R1A2C3006844).

ORCID

Not Applicable

References

- Amor R, Dimyadi J (2021) The promise of automated compliance checking. *Developments in the Built Environment* 5:100039, DOI: 10.1016/j.dibe.2020.100039
- Autodesk I (2022a) Autodesk revit. Retrieved May 18, 2022, <https://www.autodesk.co.kr/products/revit>
- Autodesk I (2022b) Navisworks. Retrieved May 18, 2022, <https://www.autodesk.co.kr/products/navisworks>
- Beckett D, Berners-Lee T, Prud'hommeaux E and Carothers G (2022) RDF 1.1 Turtle. Retrieved May 18, 2022, <http://www.w3.org/TR/2014/REC-turtle-20140225/>
- Bentley Systems I (2022) Bentley systems. Retrieved May 18, 2022,

- <https://www.bentley.com/>
- Cronembold JR, Law KH (1988) Automated processing of design standards. *Journal of Computing in Civil Engineering* 2(3):255-273, DOI: 10.1061/(ASCE)0887-3801(1988)2:3(255)
- Czmoch I, Pękala A (2014) Traditional design versus BIM based design. *Procedia Engineering* 91:210-215, DOI: 10.1016/j.proeng.2014.12.048
- Dimyadi J, Amor R (2013) Automated building code compliance checking—where is it at. 19th International CIB World Building Congress, May 5-9 Brisbane, Australia
- Dimyadi J, Clifton C, Spearpoint M, Amor R (2016) Computerizing regulatory knowledge for building engineering design. *Journal of Computing in Civil Engineering* 30(5):C4016001, DOI: 10.1061/(ASCE)JCP.1943-5487.0000572
- Eastman CM (1981) Database facilities for engineering design. Proceedings of the IEEE 69(10):1249-1263, DOI: 10.1109/PROC.1981.12165
- Eastman C (2009) Automated assessment of early concept designs. *Architectural Design* 79(2):52-57, DOI: 10.1002/ad.851
- Eastman C, Lee JM, Jeong YS, Lee JK (2009) Automatic rule-based checking of building designs. *Automation in Construction* 18(8):1011-1033, DOI: 10.1016/j.autcon.2009.07.002
- Fenves SJ (1966) Tabular decision logic for structural design. *Journal of the Structural Division* 92(6):473-490, DOI: 10.1061/JSDEAG0001567
- Fenves SJ, Garrett J, Kiliccote H, Law K, Reed K (1995) Computer representations of design standards and building codes: US perspective. *The International Journal of Construction Information Technology* 3(1):13-34
- Fenves SJ, Slava MT, Barnett JP (1987) SASE-Standards analysis, synthesis, and expression program: User Manual. National Bureau of Standards, Gaithersburg, MD, USA
- Garrett JH, Fenves SJ (1987) A knowledge-based standards processor for structural component design. *Engineering with Computers* 2:219-238
- Horridge M, Bechhofer S (2011) The owl api: A java api for owl ontologies. *Semantic web* 2(1):11-21, DOI: 10.3233/SW-2011-0025
- Horrocks I, Patel-Schneider PF, Boley H, Tabet S, Grosz B, Dean M (2022) SWRL: A semantic web rule language combining OWL and RuleML. Retrieved May 18, 2022, <https://www.w3.org/submissions/2004/SUBM-SWRL-20040521/>
- Jia J, Gao J, Wang W, Ma L, Li J, Zhang Z (2022) An automatic generation method of finite element model based on BIM and Ontology. *Buildings* 12(11):1949, DOI: 10.3390/buildings12111949
- Jiang L, Shi J, Wang C (2022) Multi-ontology fusion and rule development to facilitate automated code compliance checking using BIM and rule-based reasoning. *Advanced Engineering Informatics* 51:101449, DOI: 10.1016/j.aei.2021.101449
- Jiang S, Wang N, Wu J (2018) Combining BIM and ontology to facilitate intelligent green building evaluation. *Journal of Computing in Civil Engineering* 32(5):04018039, DOI: 10.1061/(ASCE)JCP.1943-5487.0000786
- Johannes D, Pauwels P, Amor R (2016) Modelling and accessing regulatory knowledge for computer-assisted compliance audit. *Journal of Information Technology in Construction (ITcon)* 21:317-336
- KCI (2012) Korea Structural Concrete Design Code 2012
- KCSC (2023) Construction standard code - structural design code. Korea Construction Standards Center (KCSC). Retrieved December 18, 2023, <https://www.kcsc.re.kr/Search/ListCodes/101014>
- Khattra SK, Singh J, Rai HS (2022) A statistical review to study the structural stability of buildings using building information modelling. *Archives of Computational Methods in Engineering* 29(5):2857-2874, DOI: 10.1007/s11831-021-09677-5
- Kim Y, Chin S, Choo S (2022) BIM data requirements for 2D deliverables in construction documentation. *Automation in Construction* 140:104340, DOI: 10.1016/j.autcon.2022.104340
- KMOLIT (2016) Korean highway bridge design code (Limit State Design). Korean Ministry of Land, Infrastructure and Transport, South Korea
- KMOLIT (2020) Construction Industry BIM Basic Guidelines. Korean Ministry of Land, Infrastructure and Transport, South Korea
- Kong L, Yang Q, Zhou Q, Xing J, Sun X, Zou R (2022) Embedding knowledge into BIM: A case study of extending BIM with firefighting plans. *Journal of Building Engineering* 49:103999, DOI: 10.1016/j.jobbe.2022.103999
- Leite FL (2019) BIM for design coordination: A virtual design and construction guide for designers, general contractors, and MEP subcontractors. John Wiley & Sons, Hoboken, NJ, USA, 57-83
- Malsane S, Matthews J, Lockley S, Love PE, Greenwood D (2015) Development of an object model for automated compliance checking. *Automation in Construction* 49:51-58, DOI: 10.1016/j.autcon.2014.10.004
- Noland JL, Feng CC (1971) Formulation of decision logic tables. *Journal of the Structural Division* 97(1):439-461, DOI: 10.1061/JSDEAG0002807
- O'Connor MJ, Das AK (2009) SQWRL: A query language for OWL. Proceedings of the 5th International Workshop on OWL: Experiences and Directions (OWLED 2009), October 23-24 Chantilly, VA, USA
- Park SI, Lee S-H, Almasi A, Song J-H (2020) Extended IFC-based strong form meshfree collocation analysis of a bridge structure. *Automation in Construction* 119:103364, DOI: 10.1016/j.autcon.2020.103364
- Patlakas P, Livingstone A, Hairstans R, Neighbour GB (2018) Automatic code compliance with multi-dimensional data fitting in a BIM context. *Advanced Engineering Informatics* 38:216-231, DOI: 10.1016/j.aei.2018.07.002
- Pauwels P, Terkaj W (2016) EXPRESS to OWL for construction industry: Towards a recommendable and usable ifcOWL ontology. *Automation in Construction* 63:100-133, DOI: 10.1016/j.autcon.2015.12.003
- Preidel C, Borrmann A (2016) Towards code compliance checking on the basis of a visual programming language. *Journal of Information Technology in Construction (ITcon)* 21(25):402-421
- Preidel C, Borrmann A (2018) BIM-based code compliance checking. Springer, Cham, Midtown Manhattan, NYC, USA
- Proctor M (2011) Drools: A rule engine for complex event processing. Proceedings of the 4th international conference on Applications of Graph Transformations with Industrial Relevance, October 4-7 Budapest, Hungary
- Rasdorf WJ, Lakmazaheri S (1990) Logic-based approach for modeling organization of design standards. *Journal of Computing in Civil Engineering* 4(2):102-123, DOI: 10.1061/(ASCE)0887-3801(1990)4:2(102)
- Rogers CW (2023) Structures of building design standards: Leveraging network analysis to understand perceived complexity. PhD Thesis, University of Pittsburgh, Pittsburgh, PA, USA
- Sibenik G, Kovacic I (2020) Assessment of model-based data exchange between architectural design and structural analysis. *Journal of Building Engineering* 32:101589, DOI: 10.1016/j.jobbe.2020.101589
- SkyCiv (2022) SKYCIV S3D. Retrieved May 18, 2022, <https://skyciv.com/structural-software/s3d-structural-analysis-software/>
- Solibri I (2022) Solibri model checker. Retrieved May 18, 2022, <https://www.solibri.com/>
- Solibri I (2023) Using gatekeeper rules to filter components into sub-rules. Retrieved December 18, 2023, <https://help.solibri.com/hc/en-us/articles/4415773412247-Using-Gatekeeper-Rules-to-Filter-Components-into->

Sub-Rules

- Sriram D, Maher ML, Fenves SJ (1984) Knowledge-based expert systems in structural design. *Advances and Trends in Structures and Dynamics*, October 22-25 WDC, USA
- Sydora C, Stroulia E (2020) Rule-based compliance checking and generative design for building interiors using BIM. *Automation in Construction* 120:103368, DOI: [10.1016/j.autcon.2020.103368](https://doi.org/10.1016/j.autcon.2020.103368)
- Tan X, Hammad A, Fazio P (2010) Automated code compliance checking for building envelope design. *Journal of Computing in Civil Engineering* 24(2):203-211, DOI: [10.1061/\(ASCE\)0887-3801\(2010\)24:2\(203\)](https://doi.org/10.1061/(ASCE)0887-3801(2010)24:2(203))
- Teicholz P, Sacks R, Liston K, Eastman C (2018) *Bim Handbook: A Guide to Building Information Modeling for Owners, Designers, Engineers, Contractors, and Facility Managers*. John Wiley & Sons, Hoboken, NJ, USA, 57-83
- Wu J, Zhang J (2022) Model validation using invariant signatures and logic-based inference for automated building code compliance checking. *Journal of Computing in Civil Engineering* 36(3):04022002
- Xu Z, Rao Z, Gan VJL, Ding Y, Wan C, Liu X (2019) Developing an extended IFC data schema and mesh generation framework for finite element modeling. *Advances in Civil Engineering* 2019:1-19, DOI: [10.1155/2019/1434093](https://doi.org/10.1155/2019/1434093)
- Yabuki N, Law KH (1993) An object-logic model for the representation and processing of design standards. *Engineering with Computers* 9:133-159
- Zhang J, Cui B, Gao Y, Zhang D (2023) Factors Influencing the acceptance of BIM-Based automated code compliance checking in the AEC industry in China. *Journal of Management in Engineering* 39(6):04023036, DOI: [10.1061/JMENEA.MEENG-5344](https://doi.org/10.1061/JMENEA.MEENG-5344)
- Zhang S, Boukamp F, Teizer J (2015) Ontology-based semantic modeling of construction safety knowledge: Towards automated safety planning for job hazard analysis (JHA). *Automation in Construction* 52:29-41, DOI: [10.1016/j.autcon.2015.02.005](https://doi.org/10.1016/j.autcon.2015.02.005)
- Zhang J, El-Gohary NM (2016) Semantic NLP-based information extraction from construction regulatory documents for automated compliance checking. *Journal of Computing in Civil Engineering* 30(2):04015014, DOI: [10.1061/\(ASCE\)CP.1943-5487.0000346](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000346)
- Zhang S, Teizer J, Lee J-K, Eastman CM, Venugopal M (2013) Building information modeling (BIM) and safety: Automatic safety checking of construction models and schedules. *Automation in Construction* 29:183-195, DOI: [10.1016/j.autcon.2012.05.006](https://doi.org/10.1016/j.autcon.2012.05.006)
- Zhong B, Ding L, Luo H, Zhou Y, Hu Y, Hu H (2012) Ontology-based semantic modeling of regulation constraint for automated construction quality compliance checking. *Automation in Construction* 28:58-70, DOI: [10.1016/j.autcon.2012.06.006](https://doi.org/10.1016/j.autcon.2012.06.006)
- Zhou P, El-Gohary N (2016) Ontology-based multilabel text classification of construction regulatory documents. *Journal of Computing in Civil Engineering* 30(4):04015058, DOI: [10.1061/\(ASCE\)CP.1943-5487.0000530](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000530)
- Zhou P, El-Gohary N (2018) Text and information analytics for fully automated energy code checking. *Proceedings of the 2nd GeoMEast International Congress and Exhibition on Sustainable Civil Infrastructures*, November 24-28 Cairo, Egypt
- Zhou P, El-Gohary N (2021) Semantic information alignment of BIMs to computer-interpretable regulations using ontologies and deep learning. *Advanced Engineering Informatics* 48:101239, DOI: [10.1016/j.aei.2020.101239](https://doi.org/10.1016/j.aei.2020.101239)